

Question	Answer	Mark	Mark scheme	Additional guidance
21	$16\sqrt{3} - \frac{16\pi}{3}$	P1	for identifying an angle of 60 or 120	Does not need to be in simplest form
		P1	for process to find the area of a sector of angle 60 or 120 eg $\pi 4^2 \times \frac{60}{360} (= \frac{8\pi}{3})$ or $\pi 4^2 \times \frac{120}{360} (= \frac{16\pi}{3})$	
		P1	for process to find the area of an equilateral triangle eg $\frac{1}{2} \times 4 \times 4 \times \sin 60 (= 4\sqrt{3})$ or $\frac{4 \times \sqrt{4^2 - 2^2}}{2} (= 2\sqrt{12} \text{ or } 4\sqrt{3})$ or the area of an isosceles triangle or area of a right-angled triangle eg $\frac{1}{2} \times 4 \times 4 \times \sin 120 (= 4\sqrt{3})$ or $\frac{2 \times \sqrt{4^2 - 2^2}}{2} (= \sqrt{12} \text{ or } 2\sqrt{3})$	
		P1	for using area of sector – area of triangle to find area of a segment eg $\pi 4^2 \times \frac{60}{360} - \frac{1}{2} \times 4 \times 4 \times \sin 60 (= \frac{8\pi}{3} - 4\sqrt{3})$ or $\pi 4^2 \times \frac{120}{360} - \frac{1}{2} \times 4 \times 4 \times \sin 120 (= \frac{16\pi}{3} - 4\sqrt{3})$	
		A1	for $16\pi - 4(\frac{16\pi}{6} - 4\sqrt{3} + \frac{16\pi}{6})$ or $16\sqrt{3} - \frac{16\pi}{3}$ oe	